

Candidate Athila recombination events

Total candidate elements: 161

intra-Athila (between ATHILA families): 128 (2 full-5)

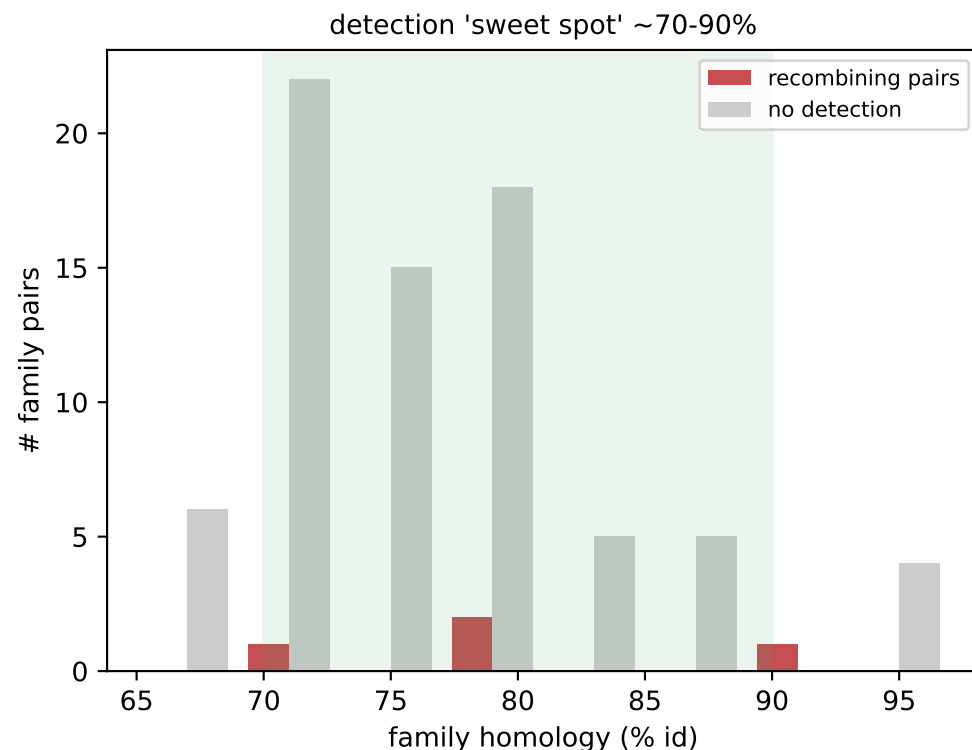
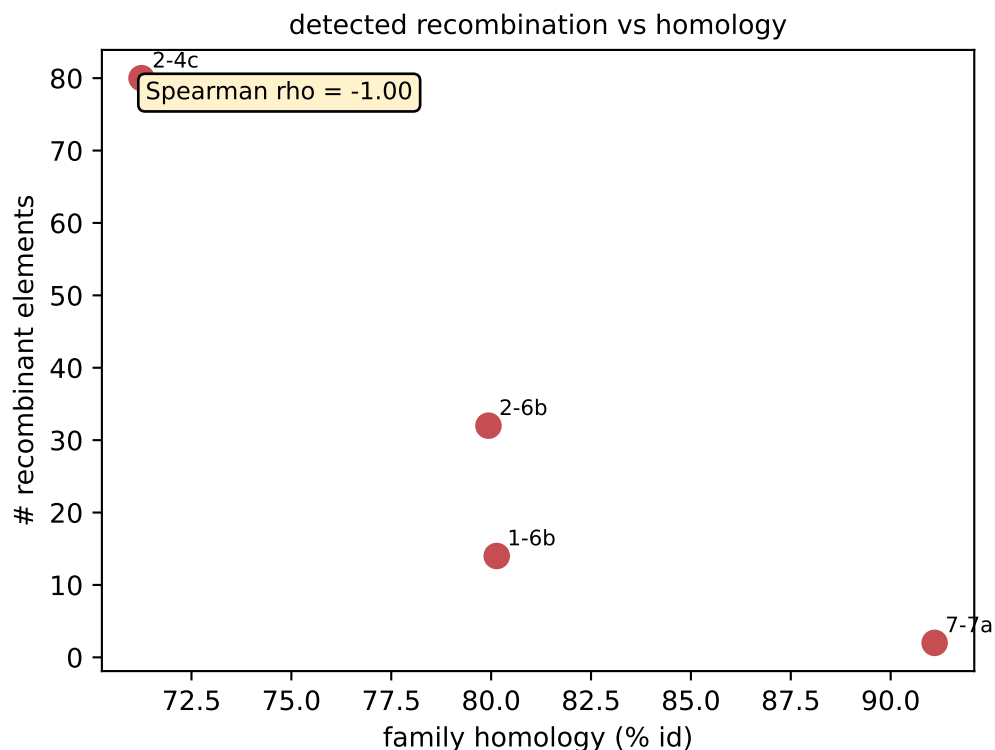
inter-clade (Athila x other LTR retro family): 33

Recombination sits at the element ENDS (GAG / INT) in every event type; the RT/RH core never swaps.

Confidence: intra-Athila = nucleotide per-domain family typing (margin $\geq 1.3\times$); inter-clade = TESorter per-domain clade call
These are CANDIDATES (not yet Phase-6 nesting-filtered or Phase-7 breakpoint-confirmed).

category	partners	swapped	# elem
intra_athila	ATHILA2,ATHILA4c	GAG	80
intra_athila	ATHILA2,ATHILA6b	GAG	19
inter_clade	Athila+CRM	CHDCR	16
inter_clade	Athila+Retand	INT;aRH	13
intra_athila	ATHILA2,ATHILA6b	INT	13
intra_athila	ATHILA1,ATHILA6b	INT	11
intra_athila	ATHILA1,ATHILA6b	GAG	3
inter_clade	Athila+Ivana	INT	2
intra_athila	ATHILA7,ATHILA7a	GAG	2
inter_clade	Athila+CRM	INT	1
inter_clade	Athila+Tork	GAG	1

Recombination bias vs family homology (Robin's question)

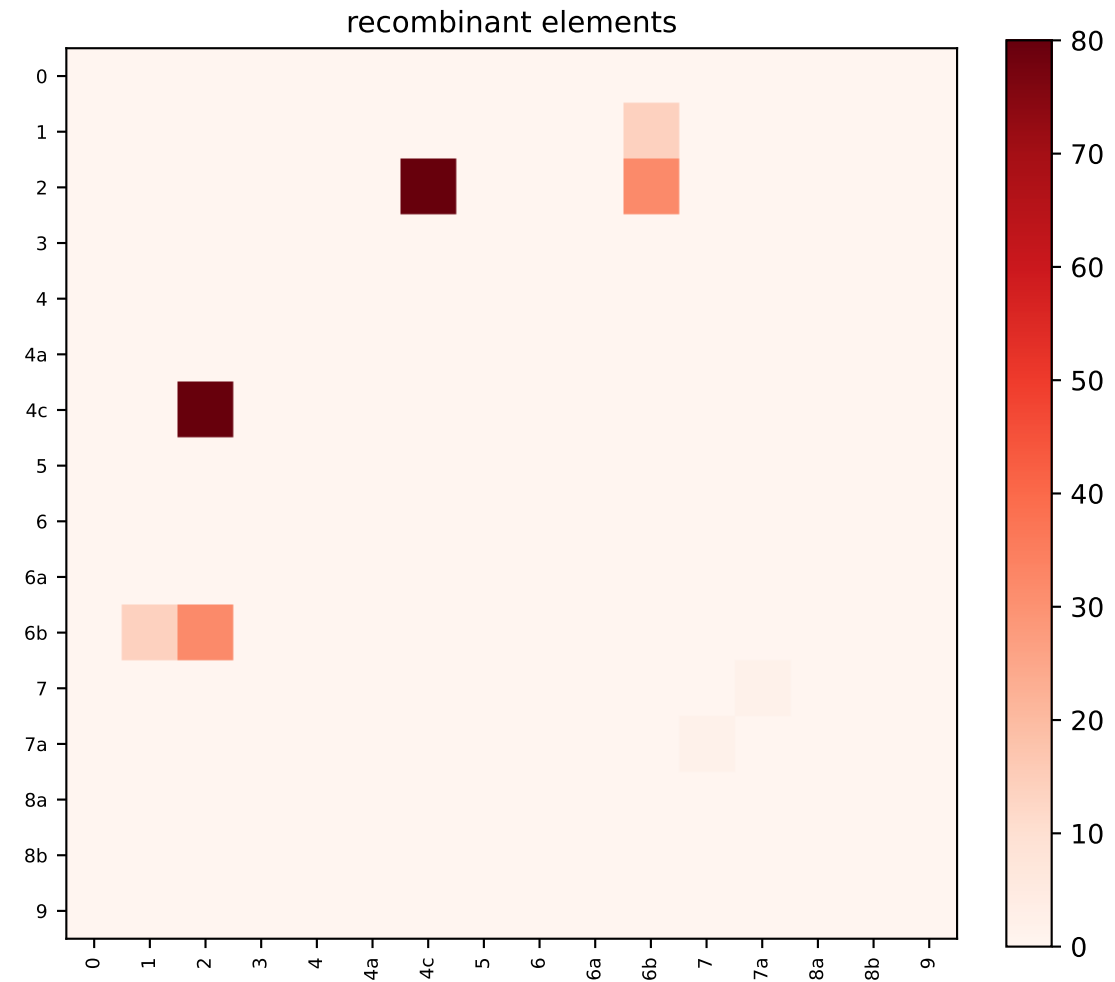
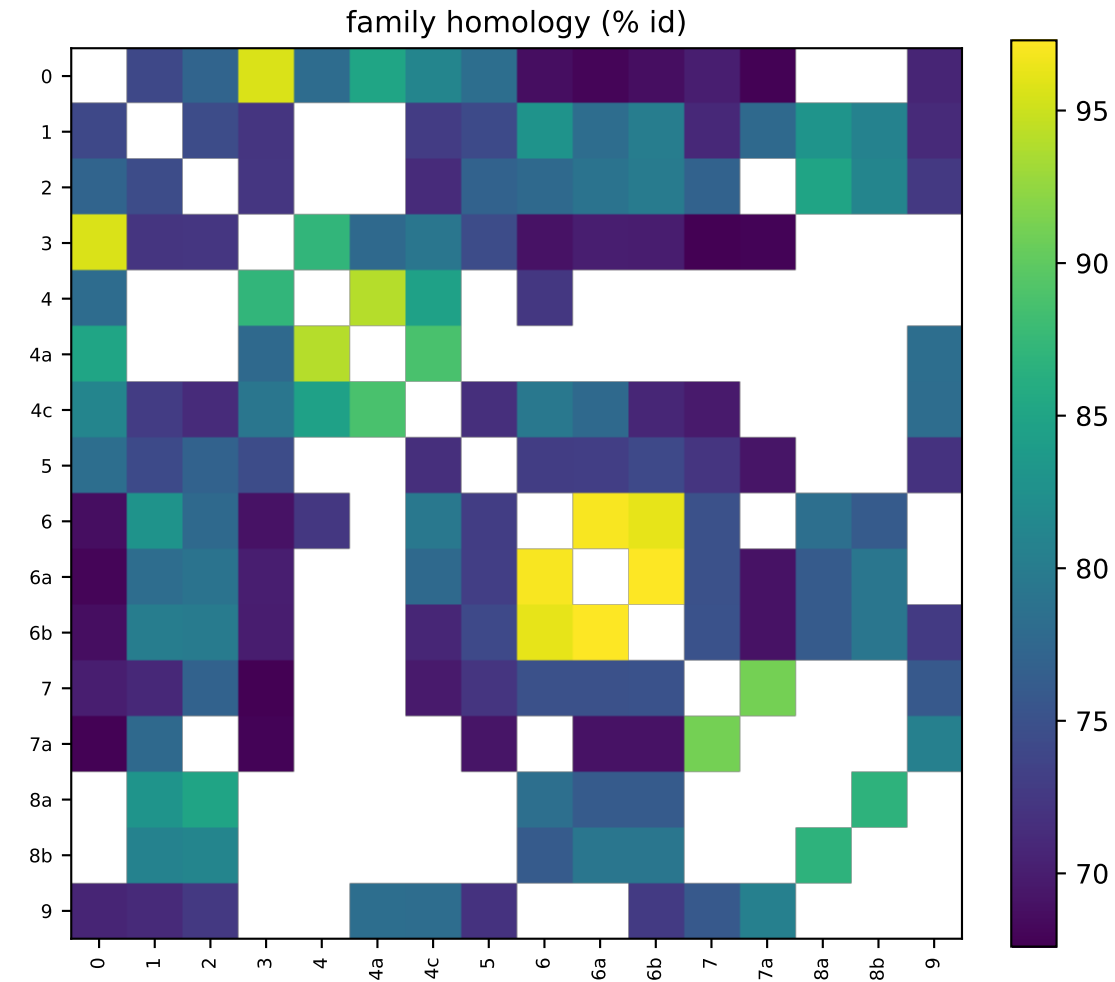


ANSWER (with the essential caveat):

- Raw detected recombination is **NEGATIVELY** correlated with homology ($\rho = -1.0$ over the 4 detectable pairs), and the **MOST** homologous families (>93% id: ATHILA6a-6b, ATHILA0-3, ATHILA4-4a) show **ZERO** detections.
- This is largely a **DETECTION ARTEFACT**, not biology: when two families are near-identical, a swapped domain cannot be confidently assigned to one vs the other, so homologous-recombination between the most similar families is **INVISIBLE** to this method. Detection peaks at intermediate homology (~70-90%).
- Counts are also confounded by family **ABUNDANCE** (ATHILA2/4c/6b are common, so partner more often by chance).

To test the homology-bias hypothesis properly we need a method that does **NOT** rely on discriminating the swapped family: nucleotide breakpoint detection (GARD/RDP5) or diagnostic-SNP mosaic mapping, normalised by pairwise family abundance. The current data can only say recombination is readily detected between moderately-divergent Athila families.

Family homology vs recombination matrix



All candidate recombination events (rows 1-42 of 161)

category	accession	chrom	start	architecture	event	full5
inter_clade	ragtagCas-6_ColChr1to5	Chr3	17625690	GAG:Athila PROT:Athila	INT=CRM	0
inter_clade	ASS3_AT	Chr4	10134551	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	Etna-2.ragtag_Ch1to5	Chr4	10367893	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	Etna-2_kang_ragtag_Ch1to5	Chr4	10368859	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	Rubezhnoe-1_fixed_Ch1to5	Chr4	10470607	GAG:Tork RT:Athila RH:A	GAG=Tork	0
inter_clade	ragtagMed-0_ColChr1to5	Chr4	8305988	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	ragtagBELC-C-10_ColChr1to5	Chr4	9092803	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	ragtagCvi-0_ColChr1to5	Chr4	9092838	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	BELC-C-12.ragtag_Ch1to5	Chr4	9099461	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	ragtag9762_ColChr1to5	Chr4	9344328	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	_Anz-0_ColChr1to5ragtag	Chr4	9493670	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	Sf-2.ragtag_Ch1to5	Chr4	9600249	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	AS530_AT	Chr4	9832191	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	Tibet-0.ragtag_Ch1to5	Chr5	11312710	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	Got-22.ragtag_Ch1to5	Chr5	11946063	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	LI-OF-095.ragtag_Ch1to5	Chr5	12487630	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	Hs-0.ragtag_Ch1to5	Chr5	12674576	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	_6184_Ch1to5	Chr5	13239502	GAG:Athila PROT:Athila	CHDCR=CRM	0
inter_clade	Rab-R1.ragtag_Ch1to5	Chr5	13594790	GAG:Athila PROT:Athila	INT=Ivana	0
inter_clade	22005.ragtag_Ch1to5	Chr5	13595757	GAG:Athila PROT:Athila	INT=Ivana	0
inter_clade	Kelsterbach-2.ragtag_Ch1to5	Chr5	15393193	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	Dra-2.ragtag_Ch1to5	Chr5	15739333	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	Est-0.ragtag_Ch1to5	Chr5	15915508	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	_No-0_ColChr1to5ragtag	Chr5	16146685	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	St-0_kang_ragtag_Ch1to5	Chr5	16162089	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	9994.ragtag_Ch1to5	Chr5	16533472	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	ragtagMed-3_ColChr1to5	Chr5	16874362	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	ragtagERR8705874_ColChr1to5	Chr5	17027210	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	Nz-1.ragtag_Ch1to5	Chr5	17135180	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	Stw-0_Ch1to5_2026	Chr5	17311503	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	LL-0.ragtag_Ch1to5	Chr5	17812374	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	ragtagBARA-C-5_ColChr1to5	Chr5	17842294	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
inter_clade	ragtagMr-0_ColChr1to5r	Chr5	18674618	GAG:Athila PROT:Athila	INT=Retand;aRH=Retar	0
intra_athila	Moh-0.ragtag_Ch1to5	Chr1	18706603	GAG:ATHILA1;INT:ATHILA1	GAG > ATHILA1;ATHILA1 > GAG	0

All candidate recombination events (rows 43-84 of 161)

category	accession	chrom	start	architecture	event	full5
intra_athila	Sij-2.ragtag_Ch1to5	Chr3	16472303	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	AH-7.ragtag_Ch1to5	Chr3	16538225	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagEvs-12_ColChr1to5	Chr3	16599339	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Kondara.ragtag_Ch1to5	Chr3	16662274	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Sorbo.ragtag_Ch1to5	Chr3	16668016	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagHum-4_ColChr1to5	Chr3	16689894	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtag7143_ColChr1to5	Chr3	16707844	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Kar-1.ragtag_Ch1to5	Chr3	16752297	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Nov-2.ragtag_Ch1to5	Chr3	16774580	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Sus-1.ragtag_Ch1to5	Chr3	16813478	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Cdm-0.ragtag_Ch1to5	Chr3	16888913	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagSan-9_ColChr1to5	Chr3	17010882	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Sha.ragtag_Ch1to5	Chr3	17031302	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagMr-0_ColChr1to5	Chr3	17058671	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Elh-2.ragtag_Ch1to5	Chr3	17059426	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	_Geg-14_ColChr1to5rag	Chr3	17076446	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	LI-OF-095.ragtag_Ch1to5	Chr3	17237601	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtag9852_Ch1to5	Chr3	17299200	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Sf-2.ragtag_Ch1to5	Chr3	17331321	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ws-4.ragtag_Ch1to5	Chr3	17423380	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Per-1.ragtag_Ch1to5	Chr3	17539770	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagCAMA-C-2_ColChr1to5	Chr3	17560455	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagCAMA-C-9_ColChr1to5	Chr3	17564115	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Belmonte-4-94.ragtag_Ch1to5	Chr3	17623601	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtag9879_ColChr1to5	Chr3	17655039	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagFERR-A-8_ColChr1to5	Chr3	17665472	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagFERR-A-12_ColChr1to5	Chr3	17676692	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	9994.ragtag_Ch1to5	Chr3	17724292	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Est-0.ragtag_Ch1to5	Chr3	17736003	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	_6024_Ch1to5	Chr3	17776000	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Hs-0.ragtag_Ch1to5	Chr3	17811760	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagCat-0_ColChr1to5	Chr3	17837887	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtag9336_ColChr1to5	Chr3	17851800	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Kac-1.ragtag_Ch1to5	Chr3	17867002	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0

All candidate recombination events (rows 85-126 of 161)

category	accession	chrom	start	architecture	event	full5
intra_athila	Dra-2.ragtag_Chrlto5	Chr3	18017909	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtag9900_ColChr1to5	Chr3	18030148	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagMet-6_ColChr1to5	Chr3	18038631	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Bor-1.ragtag_Chrlto5	Chr3	18061664	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ANGE-B-10.ragtag_Chrlto5	Chr3	18061875	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagBANI-C-12_ColChr1to5	Chr3	18071314	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagANGE-B-2_ColChr1to5	Chr3	18073402	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagCas-6_ColChr1to5	Chr3	18098240	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Elk-1.ragtag_Chrlto5	Chr3	18098377	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagMERE-A-13_ColChr1to5	Chr3	18116398	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Ishikawa.ragtag_Chrlto5	Chr3	18204499	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Hiroshima.ragtag_Chrlto5	Chr3	18213796	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Sah-0.ragtag_Chrlto5	Chr3	18304635	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Abd-0.ragtag_Chrlto5	Chr3	18366754	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagERR8705874_ColChr1to5	Chr3	18428854	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	LL-0.ragtag_Chrlto5	Chr3	18466328	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Qar-8a.ragtag_Chrlto5	Chr3	18535407	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Yilong-0.ragtag_Chrlto5	Chr3	18556230	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Co-4.ragtag_Chrlto5	Chr3	18576738	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Tul-0.ragtag_Chrlto5	Chr3	18593259	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagHum-2_ColChr1to5	Chr3	18665765	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	RRS10.ragtag_Chrlto5	Chr3	18725297	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Ket-10.ragtag_Chrlto5	Chr3	18848402	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	100621.Chr_scaffolds	Chr3	18892645	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagMdc-14_ColChr1to5	Chr3	18940581	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	9806.ragtag_Chrlto5	Chr3	18941716	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagHom-0_ColChr1to5	Chr3	19133772	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagCas-0_ColChr1to5	Chr3	19208025	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Tanz_merc.ragtag_Chrlto5	Chr3	19545659	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	Tanz_wlod.ragtag_Chrlto5	Chr3	19760968	GAG:ATHILA2;RT:ATHILA	GAG->ATHILA2,ATHILA4	0
intra_athila	ragtagBELC-C-10_ColChr1to5	Chr4	10008788	GAG:ATHILA2;INT:ATHILA	GAG->ATHILA2,ATHILA6	0
intra_athila	BELC-C-12.ragtag_Chrlto5	Chr4	10015446	GAG:ATHILA2;INT:ATHILA	GAG->ATHILA2,ATHILA6	0
intra_athila	ragtagCvi-0_ColChr1to5	Chr4	10121096	GAG:ATHILA2;INT:ATHILA	INT->ATHILA2,ATHILA6b	0
intra_athila	Sf2.ragtag_Chrlto5	Chr4	10508670	GAG:ATHILA2;INT:ATHILA	GAG->ATHILA2,ATHILA6	0

All candidate recombination events (rows 127-161 of 161)

category	accession	chrom	start	architecture	event	full5
intra_athila	LL-0.ragtag_Ch1to5	Chr4	8173043	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	Mammo-1.ragtag_Ch1t	Chr4	8197597	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	Kar-1.ragtag_Ch1to5	Chr4	8218560	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	_No-0_ColChr1to5ragtag	Chr4	8277871	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	ragtagMr-0_ColChr1to5r	Chr4	8292674	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	ragtagCas-0_ColChr1to5	Chr4	8347855	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	ragtagBANI-C-12_ColCh	Chr4	8509799	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	_Dog-4_ColChr1to5ragta	Chr4	8613500	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	Got-22.ragtag_Ch1to5	Chr4	8736316	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	ragtagMed-0_ColChr1to	Chr4	8864452	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	_6184_Ch1to5	Chr4	8940665	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	100623.Chr_scaffolds	Chr4	8951130	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	Meh-0.ragtag_Ch1to5	Chr4	9131654	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	ragtagAlo-19_ColChr1to	Chr4	9157877	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	ragtagMdc-14_ColChr1t	Chr4	9212033	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	Pu2-23_ragtagChr1to5	Chr4	9247597	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	ragtagBARC-A-12_ColCh	Chr4	9347366	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	ragtagBARC-A-17_ColCh	Chr4	9380767	GAG:ATHILA2;INT:ATHIL	INT->ATHILA2,ATHILA6b	0
intra_athila	_6024_Ch1to5	Chr4	9583005	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	ragtagMos-9_ColChr1to	Chr4	9781497	GAG:ATHILA2;INT:ATHIL	GAG->ATHILA2,ATHILA6	0
intra_athila	Got-22.ragtag_Ch1to5	Chr5	11832510	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	Sus-1.ragtag_Ch1to5	Chr5	12190663	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	ragtagBANI-C-12_ColCh	Chr5	12231143	GAG:ATHILA1;INT:ATHIL	GAG->ATHILA1,ATHILA6	0
intra_athila	LI-OF-095.ragtag_Ch1t	Chr5	12373227	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	Hs-0.ragtag_Ch1to5	Chr5	12559129	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	_Altai-5_ColChr1to5ragt	Chr5	12649958	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	Meh-0.ragtag_Ch1to5	Chr5	12729901	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	Taz-0.ragtag_Ch1to5	Chr5	12920487	GAG:ATHILA1;INT:ATHIL	GAG->ATHILA1,ATHILA6	0
intra_athila	Sorbo.ragtag_Ch1to5	Chr5	12999141	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	_6184_Ch1to5	Chr5	13133224	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	Kondara.ragtag_Ch1to5	Chr5	13427325	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	Kar-1.ragtag_Ch1to5	Chr5	13433921	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	Kyr-1.ragtag_Ch1to5	Chr5	13711584	GAG:ATHILA1;INT:ATHIL	INT->ATHILA1,ATHILA6b	0
intra_athila	Toufl-1.ragtag_Ch1to5	Chr5	18230639	GAG:ATHILA7a;PT:ATHIL	GAG->ATHILA7,ATHILA7	1

Robin's question — recombination vs family homology (k-mer method)

METHOD

Family-specific k-mer markers (k=31) built with KMC from PURE elements of each ATHILA family (k-mers in $\geq 25\%$ of members), minus anything seen in any other family (kmc_tools kmers_subtract). ATHILA7/8a/8b: exemplar markers (no pure elements). Each element scanned over its INTERNAL domain span (GAG start -> INT end); a spatial switch between two families' markers = recombinant mosaic. Validation: recovers all 80 Chr3 GAG:ATHILA2/pol:ATHILA4c elements and the full-5 Toufl-1 / Elh-2 (GAG 7a > pol 7). 619 mosaics / 18,610 typed (3.3%).

RESULT (91 family pairs, zeros included)

Spearman rho with homology, permutation p (5,000 shuffles):

raw mosaics	+0.35	p=0.0014	
distinct loci	+0.35	p=0.0014	(collapses shared insertions)
per opportunity nA*nB	+0.35	p=0.0016	(abundance)
per rarer-family elem	+0.35	p=0.0016	
detectability-adjusted	+0.34	p=0.0018	

P(pair recombines at all), logistic, adj. abundance + detectability:

OR = 4.5 per +10% homology (95% CI 2.0-10.0), p = 0.0003

Negative-binomial rate models: same direction, p ~ 1e-6, but effect sizes have very wide CIs (sparse data) -> report direction, not magnitude.

Leave-one-family-out: rho stays +0.27 .. +0.43 (never reverses).

INTERPRETATION

More homologous ATHILA families recombine more, after normalising for abundance and detectability -> supports homology-dependent recombination (template switching / ectopic recombination favoured by similarity).

The earlier rho = -1 from whole-domain family typing was a DETECTION

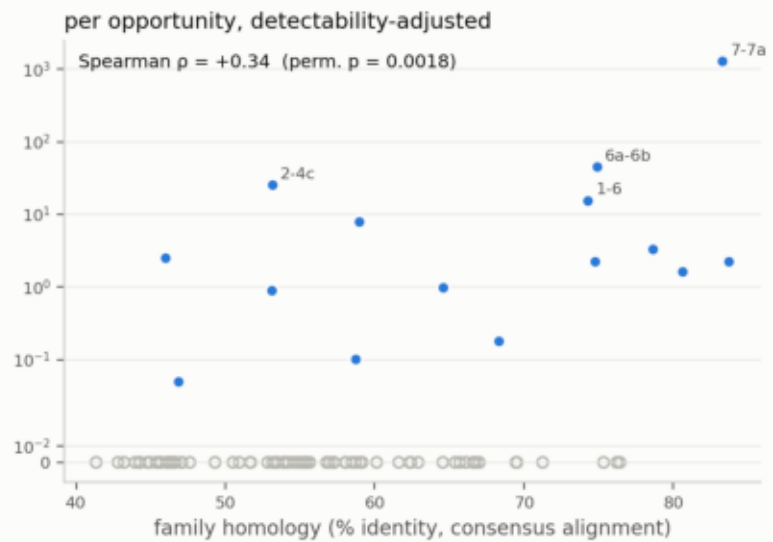
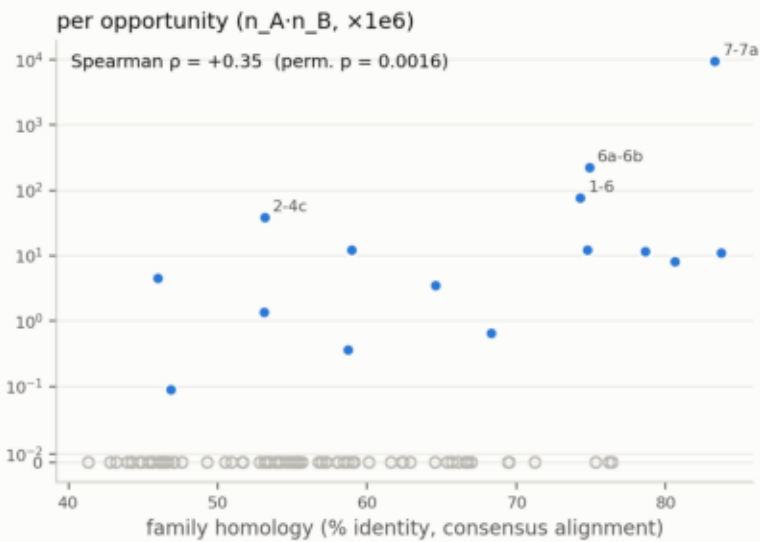
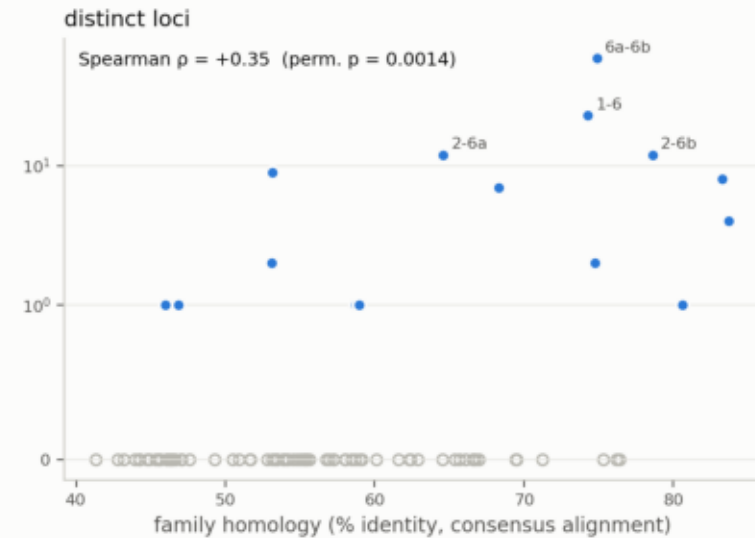
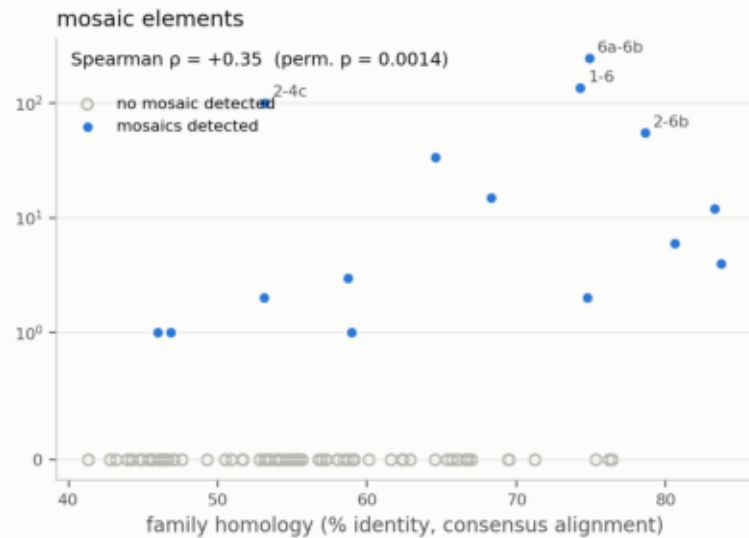
ARTEFACT: that method could not see swaps between similar families.

Notable exception: GAG:ATHILA2 / pol:ATHILA4c (53% homology) - 101 elements but only 9 distinct loci: one or a few ancestral events amplified by inheritance, not frequent low-homology recombination.

CAVEATS

- Homology = consensus-alignment identity of family exemplars (one value per pair); local identity at the breakpoint would be the more direct predictor.
- Sister families may also co-localise (shared pericentromeric niches), which raises local opportunity independently of homology.
- Calls are candidates; breakpoints not yet confirmed phylogenetically.

Does ATHILA recombination track family homology? (all family pairs, robust k-mer mosaics, internal span)



Family pairs with ≥ 1 recombinant mosaic (76 other pairs: zero)

pair	homology	raw	loci	n_a	n_b	per_opp	per_opp_det
6a-6b	74.88	246	59	1508	728	224.1	44.99
1-6	74.28	136	23	3599	490	77.12	15.58
2-4c	53.13	101	9	6415	406	38.78	25.55
2-6b	78.61	55	12	6415	728	11.78	3.292
2-6a	64.6	34	12	6415	1508	3.515	0.9823
1-2	68.3	15	7	3599	6415	0.6497	0.1816
7-7a	83.3	12	8	11	114	9569	1296
6-6a	80.6	6	1	490	1508	8.12	1.641
6-6b	83.72	4	4	490	728	11.21	2.266
2-3	58.72	3	1	6415	1294	0.3614	0.101
1-4c	53.09	2	2	3599	406	1.369	0.9017
0-3	74.74	2	2	125	1294	12.36	2.263
2-4	46.84	1	1	6415	1764	0.08837	0.04948
0-4	45.99	1	1	125	1764	4.535	2.539
4c-9	58.96	1	1	406	203	12.13	7.993